# CODEVEDA - HACKATHON 2026

- **Problem Statement Title:** AI Powered - Smart Urban Mobility & Traffic Intelligence Platform for Bengaluru
- **Theme:** Using Azure AI and real-time data to solve urban traffic congestion
- **Team Name:** Debugging Trio
- **Team Leader's Name:** Poornima H A
- **Team Members:** Muskhan B S, Monisha Thomas B, Poornima H A

## Problem Statement & Proposed Solution

- Bengaluru faces severe traffic congestion due to rapid urban growth, increasing vehicle density, and limited real-time traffic intelligence systems.
- We developed a cloud-based smart mobility platform leveraging Azure AI, geospatial analytics and real-time data to predict congestion and recommend optimized routes.
- Unique features: Dynamic signal management, Azure AI alerts, Emergency vehicle routing, 4-hour traffic prediction — all in one real-time dashboard

# TECHNOLOGY & METHODOLOGY



### React + Vite

**Frontend Framework**

Fast, modern UI with component-based architecture

### Leaflet.js

**Map Visualization**

Interactive real-time map of Bengaluru junctions

### Azure AI Services

**Cloud AI ★**

Microsoft Azure Cognitive Services for traffic analysis

### JavaScript

**Core Logic**

Signal timers, predictions, emergency detection

# FEASIBILITY AND VIABILITY



## Feasibility & Challenges

### Feasibility

A functional prototype was successfully built and demonstrated within one day using **React, Leaflet.js, and Azure AI Services.** This validates that the solution is technically feasible even with beginner-level tools and limited development time

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React Leaflet.js

### Challenges

- Integrating real-time traffic data from multiple sources
- Handling **Azure** region restrictions during development
- Ensuring scalability for deployment across 200+ traffic junctions city-wide

Azure AI Services

### Strategies

- Utilize **Azure for Students** free credits to manage cloud infrastructure costs
- Leverage **OpenStreetMap** for reliable and free geospatial data
- Plan integration of **IoT sensors** for future real-time vehicle detection and traffic analysis at junctions

# IMPACT AND BENEFITS



**30%**
Reduction in commute delays

**60s**
Emergency vehicle route clearance

**4hrs**
Advance traffic prediction window

**SDG11**
Sustainable Cities & Communities

## Future Scope

- IoT sensors at real junctions for live vehicle counting
- Integration with BMTC bus routes for multimodal routing
- City-wide deployment across all 200+ Bengaluru junctions
- Mobile app for commuters with personalized route guidance

# RESEARCH AND REFERENCES


Microsoft Azure AI Services — https://learn.microsoft.com/en-us/azure/ai-services/

Leaflet.js Map Library — https://leafletjs.com

OpenStreetMap Geospatial Data — https://www.openstreetmap.org

SDG 11 Sustainable Cities — https://sdgs.un.org/goals/goal11

Traffic Index 2024-26 — Bengaluru ranked top 10 most congested cities globally